

2.5 Numerical techniques for solving the Anderson impurity model

We have seen that approximate analytical solutions of the AIM can be obtained via the equation of motion technique. Now we want to obtain **numerical solutions**.

For convenience we define the AIM:

$$H_{\text{AIM}} = H_{\text{atom}} + H_{\text{bath}} + H_{\text{hybridization}}$$

$$U n_{\uparrow\downarrow}^c + (\epsilon_0 - \mu)(n_{\uparrow}^c + n_{\downarrow}^c)$$

interaction

$$\sum_{e\sigma} \tilde{\epsilon}_e a_{e\sigma}^\dagger a_{e\sigma}$$

$$\sum_{e\sigma} V e a_{e\sigma}^\dagger c_\sigma + V e^\dagger a_{e\sigma} c_\sigma^\dagger$$

In the EOM technique, we have seen an approximation for $G^R(\omega)$. Let us rewrite this in Matsubara frequencies:

$$G(i\omega_n) = \frac{1 - \langle n_{d\downarrow} \rangle}{i\omega_n - \epsilon_0 + \mu - \Sigma(i\omega_n)} + \frac{\langle n_{d\downarrow} \rangle}{i\omega_n - \epsilon_0 + \mu - \underline{U} - \Sigma(i\omega_n)}$$

$$\text{with } \Sigma(i\omega_n) = \sum_e \frac{|V_e|^2}{i\omega_n - \tilde{\epsilon}_e}$$

We now want to include the **interaction part** into the self-energy. However, at the same time, we want to separate the part which stems from the **hybridization**.

$\Rightarrow$ **hybridization Green**

$$\Delta(i\omega_n) = \sum_e \frac{|V_e|^2}{i\omega_n - \tilde{\epsilon}_e}$$

(formerly "self-energy" Σ)

advantages: Simple, quick, also possible on real-freq. axis, also applicable to itinerant coupling ($\Rightarrow$ see DMFT)

disadvantages: violates Luttinger theorem $\Rightarrow$ not correct out of ph-symmetry

2.5.3 Discretized Baths

These methods are not based on a perturbative expansion $\rightarrow$ represent the (infinite) degrees of freedom of the bath by a **finite number** of degrees of freedom.

Two important steps: ① discretization of bath
② Solution of finite Hamiltonian

① discretization of bath

$$\zeta_0^{-1}(i\omega_n) = i\omega_n + \mu - \epsilon_0 - \underbrace{\sum_{e=1}^{\infty} \frac{|V_e|^2}{i\omega_n - \tilde{\epsilon}_e}}_{\Delta(i\omega_n)}$$

For the next steps, it is better to work in real frequencies. The spectral theorem gives:

$$\Delta(\omega + i\eta) = \int_{-\infty}^{\infty} d\omega' \frac{\tilde{\Delta}(\omega')}{\omega - \omega' + i\eta} \quad \text{with } \tilde{\Delta}(\omega) = -\frac{1}{\pi} \text{Im} \Delta(\omega + i\eta)$$

$\rightarrow$ rewritten (like DOS) as $\tilde{\Delta}(\omega) = \sum_{e=1}^{\infty} |V_e|^2 \delta(\omega - \tilde{\epsilon}_e)$

For ∞ degrees of freedom (d.o.f.) the $(V_e, \tilde{\epsilon}_e)$ are unique.

2.5.4 other techniques (summary)

-) Hybrid-Feynman quantum Monte Carlo

discretization of time evolution (Suzuki-Trotter decomposition)

+ Hybrid-Stochastic trace estimation

$\Rightarrow$ auxiliary Ising Spins Sample by (quantum) Monte Carlo

-) Continuous-Time quantum Monte Carlo

diagrammatic expansion of partition function

$\Rightarrow$ numerically exact, state-of-the-art as of today
